

Coursework Specifications

Session : JANUARY 2025

Programme : Diploma in Information Technology (DITN)

: Diploma in Computer Science (DCSC)

Course Code : DCS1107

Course Title : Programming Fundamentals

Coursework Components: 60% (Test (10%), Lab Assessment (20%), Assignment (30%))

Test (10%)

Assessment Week: Week 8

Topics/Lectures Covered: Control structures (Sequential, Selection, Looping)

Learning Outcomes Assessed: 2

Duration: 1.5 hours

Lab Assessment (20%)

Assessment Week: Week 13

Topics/Lectures Covered: Control structures (selection, case and Looping, Function, Array, Structures)

Learning Outcomes Assessed: 1

Duration: 2 hours

Assignment (30%)

Deadline: Group Assignment (Week 14)

Learning Outcomes Assessed: 3

Marking Guidelines: 100%

Penalty for Late Submission: 1 day 10%

2 day 20%

3 day 100%

Final Examination

Format of paper:

Answer ALL **FOUR** subjective questions:

Duration: 2 hours

Module's Information:

Module	DCS1107 Programming Fundamentals	
Session	JANUARY 2025	
Programme	DITN / DCSC	
Lecturer	Chong Pui Lin	
	Email: puilin.chong@newinti.edu.my	Room: FDSIT Office Phone: 06-7982000 (ext. 2332)

Summary of Coursework Breakdown:*(as stated in course structure)*

No	Description of coursework	Learning Outcomes covered	Marks allocated
1.	TEST	LO2	10%
2.	LAB ASSESSMENT	LO1	20%
3.	ASSIGNMENT (GROUP)	LO3	30%
	CONTRIBUTION OF THE COURSEWORK TO THE COURSE		60%
	FINAL EXAM		40%
	TOTAL		100%

Penalty for late submission:

1 day – minus 20% of total mark awarded
 2 days – minus 50% of total mark awarded
 3 days – 0 mark for this piece of coursework

FINAL EXAMINATION FORMAT:**Duration: 2 hours**

Students are required to answer ALL FOUR questions. All questions carry equal marks.

Coursework #1 : TEST (10%)	
Module's Information:	
Module	DCS1107 Programming Fundamentals
Session	JANUARY 2025
Programme	DITN / DCSC
Lecturer	Chong Pui Lin
Coursework Type	TEST
Percentage	10% out of 60%
Hand-out Date	Week 8
Due Date	Week 8
Learning Outcomes Assessed:	
LO2	Assess computer programme solutions based on structured programming concepts. [C3, PLO2]

TEST Components	Coverage area	Format	Date
TEST (10%)	CONTROL STRUCTURES	- 2 Subjective questions [Total: 50 marks] <i>*Questions are subject to change</i>	Week 8

Coursework #2 – LAB ASSESSMENT (20%)	
Module's Information:	
Module	DCS1107 Programming Fundamentals
Session	JANUARY 2025
Programme	DITN / DCSC
Lecturer	Chong Pui Lin
Coursework Type	LAB ASSESSMENT
Percentage	20% out of 60%
Hand-out Date	Week 13
Due Date	Refer to the table
Learning Outcomes Assessed:	
LO1	Build a computer programme using basic logic and operation principles. [P3, PLO6]

LAB ASSESSMENT	Coverage area	Date
LAB ASSESSMENT (20%)	Control structures (selection, case and Looping) Functions / Arrays / Structures <i>*Questions are subject to change</i>	Week 13

Instructions – LAB ASSESSMENT (20%)

1. During the scheduled lab assessment, students will be given **ONE (1)** question, and they need to solve the problem by writing a program within the lab SESSION.
2. Students are required to upload a softcopy of source code (.cpp) to CANVAS 10 minutes early or before the end of lab class. [Instructions will be given later during the scheduled lab assessment]
3. Students are required to submit a softcopy of the program (which is one day after the lab assessment) where it should have the following documents:
 - a. Cover page
 - b. Marking Scheme/Rubric
 - c. Table of Content
 - d. Source code
 - e. Sample input/output screen with sample data
4. This exercise will be conducted on Week 13.
5. **No marks will be given for any late submission of the softcopy.**
6. This is an **individual** lab assessment. Students are required to complete **ALL** the question as given during the lab sessions.
7. *It is a serious offense if the students are caught cheated or copied the answer from classmates or even share their answer to the others. Zero marks will be given to the students if they are caught in doing these offenses, or this matter will be brought up to the HOP as dishonesty case.*

Guidelines:

- Your topic number will be randomly assigned in CANVAS. Go to “People” -> Lab Assessment Question

Due Date: Completed report need to be submitted by 7pm (on the SAME ASSESMENT day)

Marking Rubric

Criteria	Excellent (9-10)	Good (6-8)	Satisfactory (3-5)	Unsatisfactory (1-2)	Marks awarded
Sample Input/Output Formatting	The program works and meets all of the input/output specifications (with very creative interface design.)	The program works and produces the correct results and displays them with good interface design.	The program produces correct results but does not display them systematically or with very simple interface design.	The program is not working, with syntax error(s) and no sample output screen display.	
Readability (Program layout, Indentation and comments)	The code is exceptionally well organized and very easy to follow.	The code is fairly easy to read.	The code is readable only by someone who knows what it is supposed to be doing.	The code is poorly organized and very difficult to read.	
Correctness (Ability to code formulae and algorithms that reliably produce correct answers or appropriate results.)	Program produces correct answers or appropriate results for all inputs tested.	Program produces correct answers or appropriate results for most inputs.	Program approaches correct answers or appropriate results for most inputs, but can contain miscalculations in some cases.	Program does not produce correct answers or appropriate results for most inputs.	
Submission	The program source code and report submitted on time or before the due date.	The program source code and report submitted within a day of the due date.	The program source code and report submitted within a week of the due date.	The program source code or report submitted more than 2 weeks overdue.	
Documentation (Cover page, marking Rubric)	Clearly and effectively documented including the description of all variables, control structure, input requirements and output results.	Clearly documented including the source code, and sample input/output screens.	Basic documentation has been completed including the source code, and sample input/output screen.	No documentation included.	
				Total marks (50%)	
				Actual marks (20%)	

Coursework #3 – GROUP ASSIGNMENT (30%)

Module's Information:

Module	DCS1107 PROGRAMMING FUNDAMENTALS
Session	JANUARY 2025
Programme	DITN / DCSC (<i>Select your programme</i>)
Lecturer	Chong Pui Lin
Coursework Type	GROUP ASSIGNMENT PROGRAM + DOCUMENTATION
Percentage	30%
Hand-out Date	Week 7
Due Date	Week 14

Student's Declaration:

We declare that:

- *We understand what is meant by plagiarism.*
- *This assignment is all our own work and we have acknowledged any use of the published or unpublished works of other people.*
- *We hold a copy of this assignment which we can produce if the original is lost or damaged.*

No.	Name/Student ID	Section	Signature
1.	Chan Yan Zi	1SC1	yanzi
2.	Lee Gui Ling	1I1	guiling
3.	Nasmah Listya Yuka	1SC1	yuka
4.	Theeba Tharshini A/P Arumugam	1I1	theebe

DATE:

25.4.2025

Learning Outcomes Assessed:

LO3	Demonstrate information management and lifelong learning skills in developing new computer programmes. [A3, PLO9]
-----	---

Description of GROUP Assignment (30%)

This is a Group assignment. In this assignment you will create a **Books Store Management System** that allows a user to do the following:

- 1) Create an Admin account by supplying a user id and password.
- 2) Login using their id and password. Administrators must be able to reset their passwords if forgotten.
- 3) Quit the program.

Now if login was successful the user will be able to do the following (you can add in other functions as well):

Books Store Management System

- (i) Dashboard:
 - Upon login, the admin should be presented with a dashboard that provides an overview of the book store's operations.
 - The dashboard should include menu like sales, inventory status, and any other functions you would like to be included in the system.
- (ii) Book Management:
 - The admin should be able to add new books to the inventory.
 - The admin should be able to edit book details (title, author, price, quantity, etc.).
- (iii) Inventory Management:
 - The system should automatically update the inventory when books are sold.
 - Admins should receive notifications when stock levels are low.
 - The admin should be able to generate reports (in txt file) on the current inventory (eg: books that are required to re-order, highest sales in the bookstore).
- (iv) Order Processing:
 - Admins should have access to a list of suppliers' information.
 - The admin should be able to place orders for the books below the re-order level with the respective supplier.

The final program should include the following features:

- **File Handling:** Use file storage to maintain persistent data on items, stock levels, order, login credential and etc.
- **Input Validation:** Ensure that all inputs, such as item quantities and prices, are valid.
- **Real-time Updates:** Update product quantities immediately after an order is processed or stock is replenished.
- **Menu-driven Interface:** Provide a user-friendly interface with options for the menu and functions.

Guidelines for progress update as the following:

1. Update the Assignment Weekly Monitoring Report weekly.
2. Schedule on the assignment update as follow:

Week#	Progress Update
7	PART 1 and PART 2
8	- Start Coding: ✓ Design all the menus ✓ User input and output
	Add functions – at least 4 functions, <i>not including main() function.</i>
	Modify program by adding array of Structures
	Modify program by adding File Processing
14	Finalise the entire program and preparing for submission

ASSIGNMENT REQUIREMENTS

Part 1: Analysis

The objective of this part is to study and gather all the information about the current system running in the company. Here is the sample information that you can gather and document for this part:

1.1 Literature Review

- Choose a similar company, study the company and business background (in brief).
- Explain the system in general (what is bookstore management system) and how it is supposed to work in terms of the type of operation.
- Description of the working of the current system based on the company / business you choose (explaining from input – type of data capture, process - any formulas involve, and output – type of information produce)
- Capture sample interface of the system (from Internet) – if available

1.2 sample document

- Get a sample document from the company such as sales report / billing report. (attach into the report)
- Provide a brief description of the document.

Part 2: Design (No coding)

The objective of this part is to document your idea about the design of your proposed system (and based on the sample company). You are required to divide the system into 3 modules, such as login module, data capturing module, or output module separately.

2.1 Description of the system: explain the following using **PAC Chart** and **System Flowchart**

- Input – type of data capture
- Process – type of programming methods used, formulas need
- Output – type of output produce
- Explain the types of modules in your system (at least 3 modules/functions)

2.2 Storyboard

- Draft of sample design of your proposed system (with sample data and output/information)

PROGRAM SPECIFICATION:

You are required to create a C++ program for the system that you have analyzed and designed. Each group is required to have the based program and task distribution need to be defined. Each of the member in the group is required to have a separate program to demonstrate your own part of coding with the methods used (Function, Array, Structures, or File Processing). Then, compile the program by merging the 4 individual programs into 1 final program. Lastly, prepare the final report at the end of the assignment.

PART 1: Coding

- You need to use various types of programming methods to solve the above problem (control structures, functions, arrays and structures, File Processing)
- In order to store the data in both systems, you should use array or structures concepts.
- All input/output operations should be done in the function main and all calculations/processing in whatever other functions seem appropriate.
- You need to create a modular program that have **at least 4 functions**, and examples as the following:
 - * Heading (instructions/message)
 - * Login system
 - * Type of operations
- The program will be marked based on the following criteria:
 - * Completeness of requirements
 - * Good program practices
 - Readiness
 - Indentation
 - comments
 - * Meaningfulness in the input/output format
 - * Design/logic flow of the program
 - * Efficiency in terms of methods used
 - * Flexibility of the program
 - * Free of bugs

PART 2: REPORT REQUIREMENTS:

- Each group is required to prepare the report based on the following content arrangement:
 1. Cover Page
 2. Marking Scheme
 3. Table Of Contents
 4. PART 1
 5. PART 2
 6. Source codes
 - Group member 1 source code
 - Group member 2 source code
 - Group member 3 source code
 - Group member 4 source code
 - Final merged program
 7. Notepad files (.txt file)

8. Sample input
 9. Sample output with sample data (with proper descriptions with every screen captured)
 10. References
 11. Appendices: **(Each of you need to prepare 1 copy FOR EACH)**
 - **A – Self Reflective Report**
 - **B – Assignment Weekly Monitoring Report**
- Prepare **20 – 25 minutes** video recording to demonstrate the program execution. All members are required to participate in the video demonstration.
 - **Send the softcopy** of the source code (for example *Alan_Lee.cpp*), video recording (Google Drive) and report to CANVAS (to the folder that created) before the due date. **Any assignment will not be marked if no softcopy is received.**
 - **Submission Date: Week 14.**

End of Group Assignment

Self-Reflective Report

Name :	Chan Yan Zi
ID:	I24030164
Programme / Section :	<i>DITN (1I1) / <u>DCSC (1SC1)</u> Circle only 1</i>
Signature:	yanzi

What were you thinking and feeling when you are facing problem?

I feel like anxiety because of I scare I will delay whole group progressions. So when I facing problem I will try to fix it and debug it. If I really can't fix the problem , I will do online research to look for what's the problem that I faced and how to fix it. Beside doing online research I also asked one of my classmate who call Lin for help, to fix the problem.

Evaluate yourself on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Evaluate Gui Ling on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Evaluate Yuka on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Evaluate your group mate 3 on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Conclusion:

What else could you have done? What should you perhaps not have done? How do you feel about the team work throughout the entire assignment.

I have done login , sign up , admin , addbook , again call back function , design and after all of the member have done coding I try to compile it and debug it and make sure it run . For my group assignment , I think we have not done is calculate the book stock function because we really don't know how to code it. For this entire assignment , one of my teammate who call Theeba , last minute told us that she not going to do this assignment and we doo her part last minute. In conclusion , I don't think my group have any teamwork .

Future Enhancement:

If you are given another chance, what would you do to improve your program?

For the whole assignment I think for my part is not really as good as one of my teammate that she code, because of I not really familiar with structure , read and write file. If I got second chance I think I will improve my coding skill and help to reduce her task that she need to do .

ASSIGNMENT WEEKLY MONITORING REPORT			
Project Type		GROUP ASSIGNMENT	
Lecturer name		CHONG PUI LIN	
Student Name		Chan Yan Zi	
Student ID		I24030164	
Group member name		Nasmah Listya Yuka Lee Gui Ling	
Assignment Title		BOOKXCESS MANAGEMENT SYSTEM	
Week#	Date	Progress Update	Student Signature
7	27.2.2025	Before that we form a 4 people one group	yanzi
8	4.3.2025	We look for different book store company and discuss about which bookstore we'll be choosing	yanzi
9	11.3.2025	We decide which company we are going to do research and distribute task like company background and history	yanzi
10	18.3.2025	Done analysis part discuss which function should include in the system and sketch user interface	yanzi
11	25.3.2025	Start to code	yanzi
12	3.4.2025	Do pac and draw flowchart	yanzi
13	16.4.2026	Update pac and flowchart	yanzi
14	24.4.2025	Submit assignment	yanzi

Self-Reflective Report

Name :	Lee Gui Ling
ID:	I24027926
Programme / Section :	<u>DITN (111)</u> / DCSC (1SC1) Circle only 1
Signature:	Guiling

What were you thinking and feeling when you are facing problem?

I felt anxiety and afraid as I think I can't assist too much to my teammates as I'm not good in this course. Furthermore, I felt struggle and confused when doing the coding part, and others part, however, my teammates are assists me to solve the problems, and also my classmates too.

Evaluate yourself on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Evaluate your group mate 1 on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Evaluate your group mate 2 on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Evaluate your group mate 3 on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Conclusion:

What else could you have done? What should you perhaps not have done? How do you feel about the team work throughout the entire assignment.

In this assignment, I had done the researches, Program Analysis Chart (PAC), a small part of flow chart, video editing, and a few parts of coding. For the coding part, we missed quantity, payment method, and etc in the program. Based on this assignment's teamwork, although there was a teammate ran away, didn't took any responsibility for this assignment, but three of us work hard together and complete the tasks that the teammate not yet complete. We faced many problems at the last minutes, but we put many efforts to solve and face them. Overall, I felt very nervous and scared, but the power of cooperation allows me to face and solve problems bravely instead of being anxious.

Future Enhancement:

If you are given another chance, what would you do to improve your program?

If I have another chance in the future, I would read the assignment brief clearly before process the assignment. Moreover, I will make a checklist that needs to include for the program. Furthermore, I will arrange the time wisely to avoid conflicts with other assignments.

ASSIGNMENT WEEKLY MONITORING REPORT			
Project Type		GROUP ASSIGNMENT	
Lecturer name		CHONG PUI LIN	
Student Name		LEE GUI LING	
Student ID		I24027926	
Group member name		○ NASMAH LISTYA YUKA ○ CHAN YAN ZI	
Assignment Title		BOOKXCESS MANAGEMENT SYSTEM	
Week#	Date	Progress Update	Student Signature
7	27.2.2025	Form 4 people in a group	Ling
8	4.3.2025	Started to discuss what bookstore system we gonna conduct	Ling
9	11.3.2025	Browse research about the bookstore management system and started to doing the tasks that distributed by the group leader.	Ling
10	18.3. 2025	Completed the analysis part and discuss which function to include and sketch the program interface	Ling
11	25.3.2025	Started Coding	Ling
12	3.4.2025	Create PAC, drawing the flow chart	Ling
13	15-18.05.2025	Coding, Updating PAC, drawing flow chart	Ling
14	20-24.05.2025	Updating PAC, Flow Chart, Coding, Submit Assignment	Ling

Self-Reflective Report

Name :	Nasmah Listya Yuka
ID:	I24029614
Programme / Section :	DITN (1I1) DCSC (1SC) Circle only 1
Signature:	yuka

What were you thinking and feeling when you are facing problem?

Distressed. "What problem is happening now?" is mostly what happened internally. This couldn't be helped but I wished this assignment was not released on Ramadhan so that I can focus on my deen and have the actual energy to finish it. Most of the time I ask my friends and classmate to help me solve the problem

Evaluate yourself on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Evaluate your group mate 1 (Yan Zi) on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Evaluate your group mate 2 (Gui Ling) on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Evaluate your group mate 3 (Theeba) on the completion and achievement of the assignment. Rate from 1 – 10.

1 2 3 4 5 6 7 8 9 10

Conclusion:

What else could you have done? What should you perhaps not have done? How do you feel about the team work throughout the entire assignment.

I could have done the work earlier. I shouldn't have started the work later. The teamwork was definitely not good at the start. We had a lot of confusion and even myself was confused due to being absent since I was sick. I would say I regret getting sick that time but it was inevitable so it's very out of my control. However, I do wish the other member actually cared about the assignment as much as I care about it. We started rushing things when it was really near the deadline. To add more pressure, our other team member, Theeba, bailed out at the last minute and we had to do her part which dragged the process even more. She was not responsive. I gave her such an easy part to do and even that she didn't submit on time. I hated that.

YanZi was very much an action oriented person. But because of that she gave a lot of unreliable response when I ask her something such as deciding which bookstore website and if we should add remove item function in our program.

As for GuiLing, she was responsive in most cases and I don't have any negative comment to give. However, she did say it herself that she's weak in this subject so she needs a lot of time to understand the assignment. So because of this I was worried to give her a little hard task since it might stress her out.

Future Enhancement:

If you are given another chance, what would you do to improve your program?

I would choose to do everything that is required in the assignment brief. I would also like to make a more advanced program.

ASSIGNMENT WEEKLY MONITORING REPORT			
Project Type		GROUP ASSIGNMENT	
Lecturer name		CHONG PUI LIN	
Student Name		Nasmah Listya Yuka	
Student ID		I24029614	
Group member name		Chan Yan Zi Lee Gui Ling	
Assignment Title		BOOKXCESS MANAGEMENT SYSTEM	
Week#	Date	Progress Update	Student Signature
7		We formed a group of four as we were told by the lecturer	yuka
8	5 th March	We picked a bookstore website to be used for our assignment which is BookXcess	yuka
9	10 th March	I distributed our tasks for the analysis part	yuka
10	19 th March	We discussed about what function to put inside our program and have started designing	yuka
12	12 th April	I continued drawing the storyboard from where my groupmate left	yuka
14	20 th April	I finished making the PAC for each function (only required process and alternative solution) while the rest is done by Theeba. I listed all the functions we might need and the others picked which function they want to work on I started doing the genre list function	yuka
14	22 th April	I finished making the genre list and book list function	yuka
14	23 th April	Our teammate Theeba didn't submit her part and said it's better if others do it instead and agreed to be excluded from the presentation video so I took over her part which is checkout function and passed the cart function to Yanzi. however, halfway through she gave up bcs she couldn't understand so I also did her part and gave the compiling task to Yanzi.	yuka
14	24 th April	We rushed to finish the code, create report and record video.	yuka

DCS1107 PROGRAMMING FUNDAMENTALS

Marking Rubric

JANUARY 2025

SOURCE CODE + REPORT (50 marks)						
Criteria	Unsatisfactory	Satisfactory	Good	Excellent	Comments	Marks
Delivery (10 marks)	- Completed less than 70% of the requirements. - Not delivered on time or not in correct format (disk, email, etc.) (0-4 marks)	- Completed between 70-80% of the requirements. - Delivered on time, and in correct format (disk, email, etc.) - Basic documentation has been completed including the source code, and sample input/output screen. (5-6 marks)	- Completed between 80-90% of the requirements. - Delivered on time, and in correct format (disk, email, etc.) - Clearly documented including the source code, and sample input/output screens (7-8 marks)	- Completed between 90-100% of the requirements. - Delivered on time, and in correct format (disk, email, etc.) - Clearly and effectively documented including the description of all variables, control structure, input requirements and output results. (9-10 marks)		

Functionality + Correctness (20 marks)	- Majority of the program's functions were mainly taken from past system. - Not useful to target users as the program was not working. (0-8 marks)	- Some minor functions/features were created by the students. - Some features/functions are useful to target users. - Some features are presented in a way that clear but still contains major programming weaknesses and bad practices. (9-13 marks)	- Most of the functions and features are developed by the student with good justification. - Generally acceptable and useful to target users. - Generally the code and design is clear and understandable. Only minor bad practices are found. (14-16 marks)	- The program is entirely original. - The functions of the program are very useful to target users. - Excellent design of program code and design. (17-20 marks)		
User-friendly output – INTERFACE DESIGN (10 marks)	The program is producing incorrect results (0-4 marks)	The program produces correct results but does not display them correctly or does not follow specifications (5-6 marks)	The program produces the correct results and displays them correctly. It also meets most of the other specifications (7-8 marks)	The program meets all of the specifications required and works. (9-10 marks)		
Report Formatting (10 marks)	Work is illegible, format changes throughout, e.g. font type, size etc.	Mostly consistent format. Figures and tables are legible, but not convincing.	Format is generally consistent including heading styles and captions.	Format is generally consistent including heading styles and captions.		

	Figures and tables are sloppy and fail to provide intended information. (<2 marks)	(2- 5 marks)	Figures and tables are neatly done and provide intended information. (6 - 8 marks)	Figures and tables are neatly done and provide intended information. (9 - 10 marks)		
[A] - TOTAL MARKS (50 marks)						

INDIVIDUAL MARK : DEMONSTRATION & WEEKLY PROGRESS (50 marks)							
Criteria	Major problems	Minor problems	Good	Awarded Mark			
				Student 1	Student 2	Student 3	Student 4
Video Recording: Clarity of Explanation (15 marks)	The explanation is difficult to follow, with poor coverage of key features. Important functionality is unclear or missing from the demonstration. (0 - 5 marks)	The explanation covers the basic features but lacks depth or clarity in certain parts. Some features may not be fully explained or demonstrated. (6 - 10 marks)	The system is clearly explained with a thorough introduction, purpose, and description of each feature. The demonstration is easy to understand, with step-by-step guidance through the program's features. All major functionality is described. (10 – 15 marks)				
Visual and Audio Quality (5 marks)	The video or audio quality is subpar, with unclear visuals or frequent audio disruptions (e.g., background noise, poor volume). The presenter is	The video and audio are acceptable but may have noticeable issues (e.g., slightly blurry visuals, low volume, or minor	The video quality is clear, with well-lit visuals and easy-to-read text on the screen. The audio is crisp and clear, with no background				

	difficult to understand at times. (0 - 1 marks)	background noise). The presenter's voice is mostly clear but may be hard to understand at times. (2 - 3 marks)	noise, and the presenter's voice is easy to understand. (4 – 5 marks)				
Weekly Progress + Project Management + Each member demonstrate their contributions in the video. (30 marks)	No record or minor record of project management or monitoring with minimum contribution to the overall project. (1 - 10 marks)	Project was planned but records of modification were incomplete. Partial contribution to the overall project. (10 – 20 marks)	- Project planned effectively and managed with records of modification to project planned. - Major contribution to the overall project with leading and monitoring of the progress. (21 – 30 marks)				
[B] - TOTAL MARKS (50 marks)							

I, hereby acknowledge has received the following piece of coursework component marks.

STUDENT NAME & ID	[A]	[B]	TOTAL	
	50%	50%	100%	30%
1. <i>Chan Yan Zi</i> I24030164				
2.Lee Gui Ling I24027926				
3.Nasmah Listya Yuka I24029614				
4. Theeba Tharshini A\P Arumugam				

Contents

PART 1: ANALYSIS.....	2
1.1 Literature Review.....	2
1.2 Sample Document	5
PART 2: DESIGN (NO CODING)	6
Program Analysis Chart (PAC)	6
.....	Error! Bookmark not defined.
-	Error! Bookmark not defined.
SOURCE CODE.....	10
Main Login	Error! Bookmark not defined.
Welcome	Error! Bookmark not defined.
Login or signup	Error! Bookmark not defined.
All combined.....	Error! Bookmark not defined.
Notepad files.....	Error! Bookmark not defined.
SAMPLE INPUT OUTPUT RUN.....	Error! Bookmark not defined.

PART 1: ANALYSIS

1.1 Literature Review

i. Business background

BookXcess was founded in 2006 by husband-and-wife duo Andrew Yap and Jacqueline Ng . The idea behind the company is to sell remaindered books purchased at a discount from publishers to make books more affordable and accessible to the general public in excellent condition but sold at a fraction of the original price. The first BookXcess store opened in Amcorp Mall, Petaling Jaya, Malaysia. The concept quickly gained traction due to the attractive pricing and the wide variety of books available. Over the years, BookXcess expanded its presence across Malaysia, opening multiple outlets in shopping malls and commercial areas. The company also ventured into online sales, allowing customers to purchase books through its website. The company organizes various events, book fairs, and reading programs to encourage literacy and a love for books among all age groups. BookXcess is known for its unique and creative store designs. One of its most famous outlets is the "BookXcess Sunway Pyramid," which features a labyrinth-like design and a striking visual appeal, making it a popular destination for book lovers and Instagrammers alike. The company has also experimented with other innovative store concepts, such as combining book sales with cafes and other lifestyle elements to enhance the customer experience.

ii. Bookstore Management system

A bookstore management system is a computer program that helps a bookstore run smoothly. It keeps track of books in stock, sales, customer orders, and payments. It also helps manage suppliers and store records, making work easier for the owner and staff. With this system, a bookstore can quickly find books, check prices, and see what needs to be restocked. It saves time, reduces mistakes, and improves customer service.

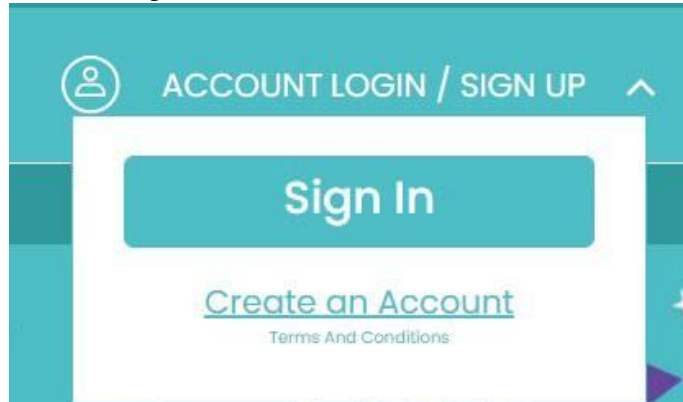
In the Bookxcess system, users need to create an account to purchase books online. If a user already has an account, they can log in right away and start shopping. Users can find books by typing the book title, interest, keyword, author or ISBN in the search bar or clicking on the categories section. Users can also click on "Best Sellers" to see the most popular books or check "Details & Promotions" to find books that are on discount. When a user clicks on a book, the system will display the book's price and details.

If a user wants to buy a book, they just need to find the book that they want and click on it. Then, they can select the number of copies they want and click the "Add to Cart" button. Users can remove or pay for the items by clicking on the "Cart" section and proceeding to checkout. During checkout, the system processes the payment and updates the inventory, making sure the stock management is accurate.

iii. Current working system

1. Input

Account login detail



Create account

Already have an account? [Login here](#)

Salutation*	First name*
Miss ▼	qwert
Last name*	
qwer	
Nationality*	
Malaysian ▼	

Search queries



Shipping address

Shipping Address		Country*	
Address Line 1*	Address Line 2*	Malaysia ▼	
1234	12345	Date of Birth	
Postcode*		Date	Month
12345		8 ▼	4 ▼
City*		Year	
asdf		2015 ▼	
State*		Mobile Number*	
Kuala Lumpur ▼		123456789	
		Email Address*	
		123456@gmail.com	
		☒ I Agree to The Terms & Conditions	
Sign up to be BookXcess member			

1. Process
2. **Search & Filtering:** The system matches user queries with the book database.
3. *Price Calculation:*
 1. Total price: $\text{quantity} \times \text{price}$
 2. If discounts apply: $\text{Final Price} = \text{Original Price} - (\text{Original Price} \times \text{Discount Rate})$
 3. If tax applies: $\text{Final Price} = \text{Subtotal} + (\text{Subtotal} \times \text{Tax Rate})$
4. *Stock Management:*
 1. When a book is purchased: $\text{New Stock} = \text{Old Stock} - \text{Quantity Sold}$
 2. If stock = 0, mark the book as "Out of Stock"
5. *Payment Processing*
 1. Card verification, applying discounts, generating invoices

2. Output

Cart summary

CART SUMMARY

Subtotal (1 item)	RM 174.75
Savings	(-RM 114.85)
Total Cart Value	RM 59.90

IN YOUR CART >



Sourdough

RM 119.80

- 2 +

ADD TO WISHLIST X REMOVE ITEM

TOTAL CART VALUE **RM 119.80**

CHECK OUT

CONTINUE SHOPPING

1.2 Sample Document



The top part of the receipt shows little information about the purchasing such as which staff was tasked at the cashier, sales number and the sales date. In the middle, it shows the items name, quantity, unit price, discounted amount and the total. At the last part, it shows the total rounding, subtotal, total discount and the net total of the product purchased.

PART 2: DESIGN (NO CODING)

Program Analysis Chart (PAC)

Login

Data Given	Required Output
name, pass	Login Successful
Required Process	Alternative Solution
Repeat the following statement until login successful: 5. User input username & password 6. Compare the entered username & password with the stored data 7. If correct, display "Login Successful" 8. If incorrect, display "Login Unsuccessful" 9. Counter update by 1	10. Define Name "qwert" 11. Define Pass "abc_1234" 12. Declare name and pass as character and user input 13. Declare failcount as integer 14. Use sequential control structure 15. Use selection control structure (if-else statement) 16. Use looping control structure (do-while loop)

Sign Up

Data Given	Required Output
choice, name, pass, conf_pass (user input)	Sign Up Complete
Required Process	Alternative Solution
Repeat following statement until password == reentered 17. User input new username & password 18. Reenter the new input for confirmation 19. If both password match, store username & password into file and display " Sign Up Complete" 20. If password doesn't match, display "Incorrect username or password. Re-enter again"	21. Declare name, pass, conf_pass as character and user input 22. Declare failcount and choice as integer 23. Use selection control structure (if-else statement) 24. Use sequential control structure 25. Use looping control structure (do-while loop)

Admin

Data Given	Required Output
choice, PASS (user input)	Category
Required Process	Alternative Solution
Repeat following statement until password correct: 26. User input PASS 27. If password match, call addbook function 28. If password doesn't match, display "invalid Option. Re-enter again"	29. Declare choice and PASS as integer and user input 30. Declare failcount = 0 31. Use selection control structure (if-else statement) 32. Use sequential control structure 33. Use looping control structure (do-while loop)

Main Dashboard

Data Given	Required Output
Choice (user input)	Bookstore Dashboard 34. Browse Book by Genre 35. View Cart 36. Checkout 37. Quit Program
Required Process	Alternative Solution
1. Repeat the following until the user enter choice <1 choice > 4: -Display Main Dashboard - user input choice -if user select browse book by genre, call genre list function -if user select view cart, call cart function - if user select checkout, call checkout function -if user select quit program, end program	1. Declare choice as integer and user input 2. Use sequential control structure 3. Use looping control structure (do-while-loop) 4. Use switch case logic structure

Genre

Data Given	Required Output
Choice (user input)	Genre list 0. Dashboard 1. Romance 2. Science Fiction 3. Thriller 4. Baking 5. Business
Required Process	Alternative Solution
Repeat following until user enter 0: 1. Display genre list 2. user input genre number 3. If input is 0, return to main menu 4. If genre number is valid: - call book list function with the selected genre number 5. If genre number is invalid, display error message	1. Declare choice as integer and user input 2. Declare genre name in array and list 2. Use Sequential control structure 3. Use looping control structure (while loop) 4. call book list function based on user selection 5. Use selection control structure (multiple if else statement)

Cart & Checkout

Data Given	Required Output
choice (user input)	Total price, cart title, cart price, cart item
Required Process	Alternative Solution
1. Repeat following until user choose to checkout/ return to main menu: - Display books stored in cart - user input selection - if user select checkout, call checkout function - if user select return to main page, go back to main page 2. if user select remove item, proceed with the following: - Display cart item with index number - ask user for item number to remove - shift remaining items in array to overwrite the selected one. - decrease cart size by 1	1. Declare totalPrice = 0 2. Declare choice as integer and user input 3. Use looping structure selection (For loop) 4. Use sequential control structure 5. Use switch case control structure 6. Use Looping control structure (while loop)

Checkout

Data Given	Required Output
Choice (user input)	Summary, booklist in cart, shipping fee, totalPrice
Required Process	Alternative Solution
1. Display checkout summary - List books in cart - Show shipping fee & total price 2. Use input if they want to move payment/ return to cart 3. If user choose proceed: - Ask user to input shipping address - Ask to confirm and pay or cancel - If confirm & pay: -> display "payment successful" -> Ask to return to main dashboard or exit program -> If return – go to the main menu -> if exit – end program -> if cancel – go back to cart 4. If user choose to return to cart - Go back to cart	1. Use looping structure control (do-while loop) 2. Use witch case to: - handle confirm/cancel - handle return to dashboard/ exit

SOURCE CODE

```
#include <iostream>
#include <iomanip>
#include <cstring>
#include <fstream>

using namespace std;

#define Name "qwert"
#define Pass "abc1234"
#define shipping 5.00

struct Genre { string name; };

struct user
{
    char name[50],pass[50],conf_pass[50];
};

struct Book
{
    string title;
    string author;
    double price;
};

struct Genre genre[10] = {
    { "Romance" },
    { "Science Fiction" },
    { "Thriller" },
    { "Baking" },
    { "Business" }
};

struct Book cart[100];
int cartCount = 0;

void wel();
void loginorsignup();
void again();
void again1();
void agian2();
void login();
void signup();
void mainlogin();
void MainDashboard();
int GenreList();
void BookList(string FileName, int genreIndex);
```

```

void Checkout();
void Payment(double FinalTotal);
void Cart();
void RemoveItem();
void Exit();
void addbook();
void admin();

main()
{
    mainlogin();
    MainDashboard();
}
void mainlogin()
{
    int choice;
    wel();
    do
    {
        loginorsignup();
        cin>>choice;

        if(choice==1)
            login();
        else if (choice==2)
            signup();
        else if (choice ==3)
            admin();
        else
            again();
    } while(choice!=1 && choice !=2 && choice !=3);
}

```

```

void MainDashboard()
{
    int choice;

    do {
        cout<<endl;
        cout<<"\t-----"<<endl;
        cout<<"\t\t\t\tBookstore Dashboard\t\t\t\t"<<endl;
        cout<<"\t-----"<<endl;
        cout<<"\t\t\t\t1. Browse book by genre\t\t\t\t"<<endl;
        cout<<"\t\t\t\t2. View cart\t\t\t\t\t"<<endl;
        cout<<"\t\t\t\t3. Checkout\t\t\t\t\t"<<endl;
        cout<<"\t\t\t\t4. Quit program\t\t\t\t\t"<<endl;
        cout<<"\t-----"<<endl;
        cout<<"\tEnter your choice: ";
        cin>>choice;
        cout<<endl;
        switch(choice)
        {
            case 1:
                while (true)
                {
                    int genreChoice = GenreList();

                    if (genreChoice == 0)
                    {
                        cout<<endl;
                        cout<<endl;
                        cout<<"\t+-----"
-----+<<endl;

                        cout<<"\t\t\t\tDirecting to main
dashboard...\t\t\t\t\n";

                        cout<<"\t+-----"
-----+<<endl;

                        cout<<endl;
                        MainDashboard();
                        break;
                    }
                    else if (genreChoice >= 1 && genreChoice <=5)
                    {
                        string filenames[10] = {
                            "romance.txt",
                            "scifi.txt",
                            "thriller.txt",
                            "baking.txt",
                            "business.txt"
                        };

                        BookList(filenames[genreChoice - 1], genreChoice -
1);
                    }
                }
            }
        }
    }
}

```

```

                                else
                                again();
                                }

        break;

    case 2:
        Cart();
        break;

    case 3:
        Checkout();
        break;

    case 4:
        Exit();
        break;

    default:
        again();
        break;
    }
} while (choice < 1 || choice > 4);
}

```

```

void wel()
{
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t\t\t\t\t\t\t\t|"<<endl;
    cout<<"\t|\t\t\t\t\tWelcome to BookXcess!\t\t\t\t|"<<endl;
    cout<<"\t|\t\t\t\t\t\t\t\t\t\t|"<<endl;
    cout<<"\t+-----+"<<endl;
}

```

```

void loginorsignup()
{
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t\t\t\t\t\t\t\t|"<<endl;
    cout<<"\t|\t 1. Login    |\t 2. Sign Up  \t|\t3.Admin\t\t|"<<endl;
    cout<<"\t|\t\t\t\t\t\t\t\t\t\t|"<<endl;
    cout<<"\t+-----+"<<endl;

    cout<<"\t\t\tEnter 1 to Login. Enter 2 to Sign Up. Enter 3 to Admin.\t\t"<<endl;
    cout<<"\t Enter your choice : ";
}

```

```

void again()
{
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t Invalid Option. Reenter again\t\t\t\t|"<<endl;
    cout<<"\t+-----+"<<endl;
}

```

```
void again1()
{
    int choice;

    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t\t\t\t\t\t\t\t|"<<endl;
    cout<<"\t|\t\t\t\t1. Try Again \t\t\t\t2. Quit \t\t|"<<endl;
    cout<<"\t|\t\t\t\t\t\t\t\t\t\t|"<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t\t\t Enter 1 to Try Again. Enter 2 to Quit.\t\t\t"<<endl;
    cout<<"\t\t\t Enter your choice : ";
    cin >> choice;

    if(choice == 1){
        login();
    }
    else if(choice == 2){
        mainlogin();
    }
}
```



```

void again3()
{
    int choice;
    cout<<"\t+-----+ "<<endl;
    cout<<"\t|\t\t\t\t\t|\t\t\t\t\t|"<<endl;
    cout<<"\t\t 1. Continue add book \t\t|\t\t2. Quit \t\t|"<<endl;
    cout<<"\t|\t\t\t\t\t|\t\t\t\t\t|"<<endl;
    cout<<"\t+-----+ "<<endl;
    cout<<"\t\t\t Enter 1 to Continue add book. Enter 2 to Quit.\t\t\t"<<endl;
    cout<<"\t\t\t Enter your choice : ";
    cin >> choice;

    if(choice == 1){
        addbook();
    }
    else if(choice == 2){
        mainlogin();
    }
}

```

```
void login()
{
    char name[50], pass[50];

    int failCount = 0;
    cout<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t\t Login\t\t\t\t|"<<endl;
    cout<<"\t+-----+"<<endl;

    do
    {
        cin.ignore();
        cout<<"\t|Username : ";cin >>name;
        cout<<"\t|Password : "; cin >>pass;

        if((strcmp(name,Name)!=0) || (strcmp(pass,Pass)!=0))
        {
            failCount++;
            cout<<"\t|Login Failed !"<<endl;
            cout<<"\t|Incorrect username or password.Re-enter again."<<endl;
        }
    }while(((strcmp(name,Name)!=0)
    || (strcmp(pass,Pass)!=0)) && failCount <3 );

    if((strcmp(name,Name)==0) && (strcmp(pass,Pass)==0))
    {
        cout<<endl;
        cout<<"\t+-----+"<<endl;
        cout<<"\t|\t\t\t\tLogin successful!\t\t\t\t|"<<endl;
        cout<<"\t+-----+"<<endl;
    }
    else if (failCount == 3)
    {
        again1();
    }
}
```

```
void signup()
{
    char name[50],pass[50],conf_pass[50];
    int failCount = 0;
    cout<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t\t Sign Up\t\t\t\t\t|\"<<endl;
    cout<<"\t+-----+"<<endl;
    cin.ignore();
    cout<<"\t|Username: ";cin>>name;
    do
    {
        cout<<"\t|Password: ";cin>>pass;
        cout<<"\t|Confirm Password: ";cin>>conf_pass;

        if (strcmp(pass,conf_pass)!=0)
        {
            failCount++;
            cout<<"\t|Sign Up Failed !\"<<endl;
            cout<<"\t|Password doesn't match. Please re-enter again.\"<<endl;
        }
    }while(strcmp(pass,conf_pass)!=0 && failCount <3);
    if((strcmp(pass,conf_pass)==0) )
    {
        cout<<endl;
        cout<<"\t+-----+"<<endl;
        cout<<"\t|\t\t\t\tSign up complete!\t\t\t\t\t|\"<<endl;
        cout<<"\t+-----+"<<endl;
    }
    else if (failCount == 3)
    {
        again2();
    }
}
```

```
int GenreList()
{
    int choice;
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t\t\tGenre List\t\t\t\t\t|"<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t\t\t\t0. Dashboard\t\t\t\t\t"<<endl;
    for (int i=0; i<5; i++)
    {
        cout<<"\t\t\t\t"<<i+1<<" . "<<genre[i].name<<"\t\t\t\t\t"<<endl;
    }
    cout<<"\t+-----+"<<endl;
    cout<<"\tEnter category: ";
    cin>> choice;

    return choice;
}

void BookList(string FileName, int genreIndex)
{
    Book books[100];
    int bookCount = 0;

    ifstream file(FileName);
    if (!file)
    {
        cout<<"\t+-----+"<<endl;
        cout << "\t|\t\t\t\tError: Cannot open file '" << FileName <<"'\t\t\t\n";
        cout<<"\t+-----+"<<endl;
        return;
    }

    while (getline(file, books[bookCount].title) && getline(file, books[bookCount].author) &&
file >> books[bookCount].price)
    {
        file.ignore();
        bookCount++;
    }
    file.close();

    char again = 'Y';
    while (again == 'Y' || again == 'y')
    {
        cout<<"\n";
        cout<<"\t+-----+"<<endl;
        cout<<"\t\t\t\t"<<genre[genreIndex].name<<"Books\t\t\t\t\n";
        cout<<"\t+-----+"<<endl;

        for(int i=0; i<bookCount; i++)
        {
            cout<< "\t\t\t\t" << i+1 << " . \t" << books[i].title << endl;
```

```

        cout<< "\t\t\t\tAuthor : "<<books[i].author<<endl;
        cout<< "\t\t\t\tPrice : RM "<< books[i].price<<endl;
        cout<< "\t+-----"
+"<<endl;
    }
    cout<< "\t\t\t"<<bookCount + 1 <<" Go back to genre list\n";
    cout<< "\t+-----+"<<endl;
    cout<<endl;
    int bookChoice;
    cout<<endl;
    cout<< "\t\t\t\tEnter the book number to add to cart: ";
    cin>>bookChoice;
    cout<<endl;
    if(bookChoice == bookCount + 1)
        return;
    else if(bookChoice >=1 && bookChoice <= bookCount)
    {
        cout<< "\t+-----"
+"<<endl;

        cout<<endl;
        cout<< "\t\t\t"<<books[bookChoice - 1].title<<" added to cart!\t\t\t\n";
        cout<<endl;
        cout<< "\t+-----"
+"<<endl;

        cart[cartCount] = books[bookChoice - 1];
        cartCount++;
        cout<<endl;
        cout<< "\t\t\t\tAdd another book from the same genre? (Y/N): ";
        cin>>again;
        cout<<endl;
    }
    else{

        cout<<endl;
        cout<< "\t+-----"
+"<<endl;

        cout<< "\t\t\t\t Invalid Option. Reenter again\t\t\t\t"<<endl;
        cout<< "\t+-----"
+"<<endl;

        cout<<endl;
    }
}
return;
}

```

[illegible]

```
    Cart();  
    return;  
    break;
```

```
default:  
    again();  
    break;
```

```
}
```

```
}
```



```
void Payment(double FinalTotal)
{
    string fullname, address, city;
    int phonenum, postcode, action;
    char choice;

    cout<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t\t Checkout\t\t\t\t|"<<endl;
    cout<<"\t|\t\t\t Please enter your delivery details below.\t\t\t|"<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t\tFull Name   : ";
    cin.ignore();
    getline(cin, fullname);
    cout<<"\t\tAddress Line : ";
    getline(cin, address);
    cout<<"\t\tCity       : ";
    getline(cin, city);
    cout<<"\t\tPostcode   : ";
    cin>>postcode;
    cout<<"\t\tPhone number(without special character): ";
    cin>>phonenum;
    cout<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t\t Confirm Checkout?\t\t\t\t\t|"<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t\t\t\t\t\t\t\t\t\t\t\t\t\t\t\t|"<<endl;
    cout<<"\t|\t\t\t 1. Confirm and Pay \t\t|\t\t2. Cancel\t\t\t|"<<endl;
    cout<<"\t|\t\t\t\t\t\t\t\t\t\t\t\t\t\t\t\t|"<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\tEnter your choice: ";
    cin>>action;
    cout<<endl;
    switch(action)
    {
        case 1:
            cout<<endl;
            cout<<"\t+-----+"<<endl;
            cout<<"\t|\t\t\t\tPayment successful!\t\t\t\t\t|"<<endl;
            cout<<"\t+-----+"<<endl;
            cout<<"\tDo you want back to dashboard? ( Press 'y' to Yes / Press others to No ) : ";
            cin>>choice;
            if (choice == 'Y' || choice == 'y')
            {
                MainDashboard();
                break;
            }
            else
                Exit();
                break;
        case 2:
```

```

        cout<<endl;
        cout<<"\t+-----"

+"<<endl;

        cout<<"\t|\t\t\tReturning to checkout...\t\t\t|\t\n";
        cout<<"\t+-----"

+"<<endl;

        cout<<endl;
        Checkout();
        break;
    default:
        Cart();
        break;
    }
}

```

```
void Cart()
{
    float totalPrice = 0;
    int choice;

    if (cartCount == 0)
    {
        cout<<endl;
        cout<<"\t+-----+"<<endl;
        cout<<"\t\t\tYour cart is empty.\t\t\t\n";
        cout<<"\t+-----+"<<endl;
        cout<<endl;
        MainDashboard();
        return;
    }

    while(true)
    {
        float totalPrice=0;
        cout<<endl;
        cout<<"\t+-----+"<<endl;
        cout<<"\t\t\t\t\t Your Cart \t\t\t\t\n";
        cout<<"\t+-----+"<<endl;
        cout<<endl;

        for (int j=0; j<cartCount; j++)
        {
            cout<<"\t"<j+1<<". " <<cart[j].title<<" - RM"<<cart[j].price<<endl;
            totalPrice += cart[j].price;
        }

        cout<<"\n\tTotal: RM"<<totalPrice<<endl;
        cout<<"\t+-----+"<<endl;
        cout<<"\t\t\t1.Proceed to checkout \t\t\t\t|"<<endl;
        cout<<"\t\t\t2.Remove item \t\t\t\t\t|"<<endl;
        cout<<"\t\t\t3.Return to dashboard \t\t\t\t|"<<endl;
        cout<<"\t+-----+"<<endl;
        cout<<"\tEnter your choice: ";
        cin>>choice;

        switch(choice)
        {
            case 1:
                Checkout();
                break;
            case 2:
                RemoveItem();
                break;
            case 3:
                MainDashboard();
                break;
```

```
        default:
            again();
            Cart();
            break;
    }
}
```

```

void RemoveItem()
{
    int index;
    cout<<"\tEnter the number of the item to remove: ";
    cin>>index;

    if(index < 1 || index > cartCount)
    {
        cout<<endl;
        cout<<"\t+-----+"<<endl;
        cout<<"\t|\t|\tInvalid number. No item removed.\t|\t|\n";
        cout<<"\t+-----+"<<endl;
        cout<<endl;
        return;
    }
    double total = 0.0;
    total -= cart[index - 1].price;

    //Shift items left to overwrite the removed item
    for (int m = index-1; m<cartCount-1; m++)
    {
        cart[m] = cart[m+1];
    }

    cartCount--;
    cout<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t|\tItem removed successfully.\t|\t|\n";
    cout<<"\t+-----+"<<endl;
    cout<<endl;
}

void Exit()
{
    cout<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<"\t|\t|\tExiting program....\t|\t|\t"<<endl;
    cout<<"\t|\t|\tThank you for visiting! Goodbye.\t|\t|\t"<<endl;
    cout<<"\t+-----+"<<endl;
    cout<<endl;
    exit(0);
}

```



```

cout<<"\t\t\t Author: ";
cin>>book.author ;
cout<<"\t\t\t Price: ";
cin>>book.price ;
cout<<"\t+-----"

-----+"<<endl;

biz<<book.title<<"\n";
biz<<book.author<<"\n";
biz<<book.price<<"\n";
biz.close();
break;

case 3: cout<<"\t+-----"

--+"<<endl;

cout<<"\t\t\t Book Title: ";
cin>>book.title ;
cout<<"\t\t\t Author: ";
cin>>book.author ;
cout<<"\t\t\t Price: ";
cin>>book.price ;
cout<<"\t+-----"

-----+"<<endl;

thriller<<book.title<<"\n";
thriller<<book.author<<"\n";
thriller<<book.price<<"\n";
thriller.close();
break;

case 4: cout<<"\t+-----"

--+"<<endl;

cout<<"\t\t\t Book Title: ";
cin>>book.title ;
cout<<"\t\t\t Author: ";
cin>>book.author ;
cout<<"\t\t\t Price: ";
cin>>book.price ;
cout<<"\t+-----"

-----+"<<endl;

scifi<<book.title<<"\n";
scifi<<book.author<<"\n";
scifi<<book.price<<"\n";
scifi.close();
break;

case 5: cout<<"\t+-----"

--+"<<endl;

cout<<"\t\t\t Book Title: ";
cin>>book.title ;
cout<<"\t\t\t Author: ";
cin>>book.author ;
cout<<"\t\t\t Price: ";

```

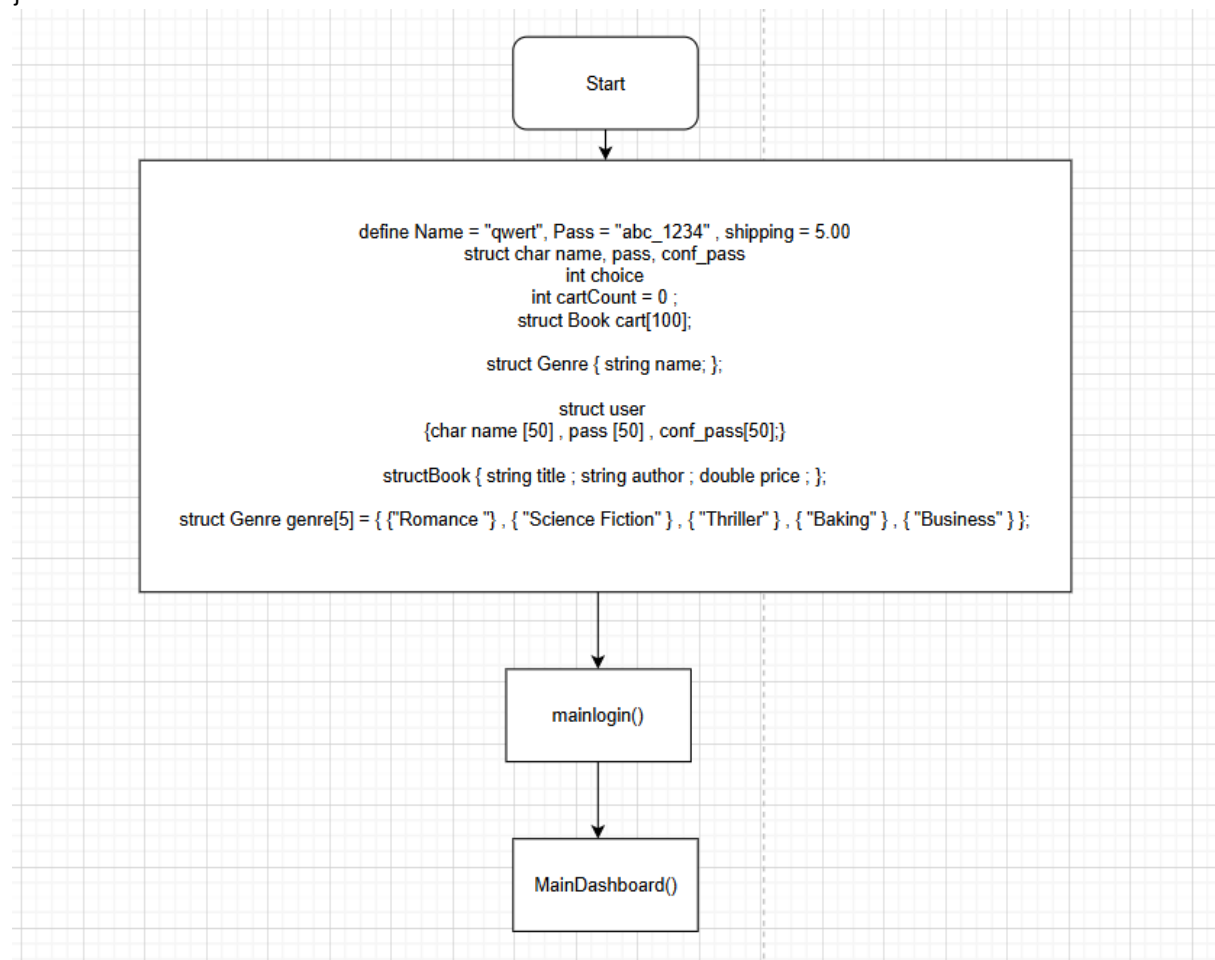
```

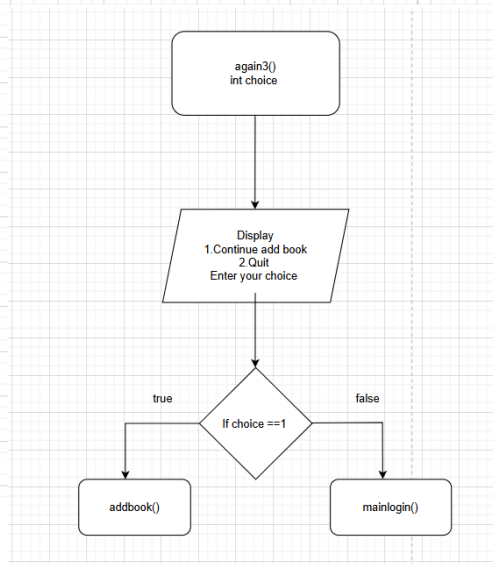
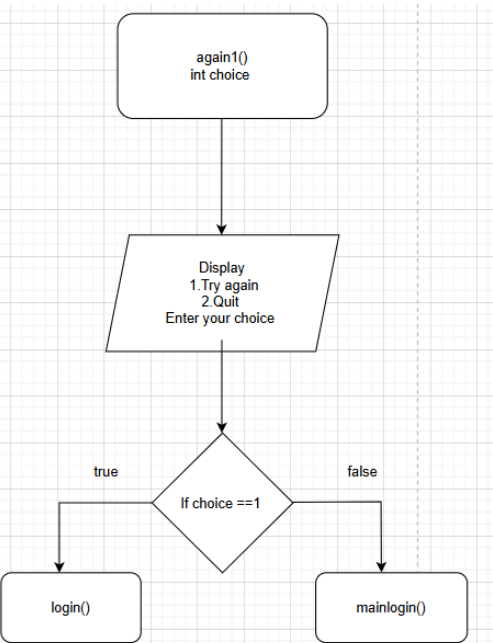
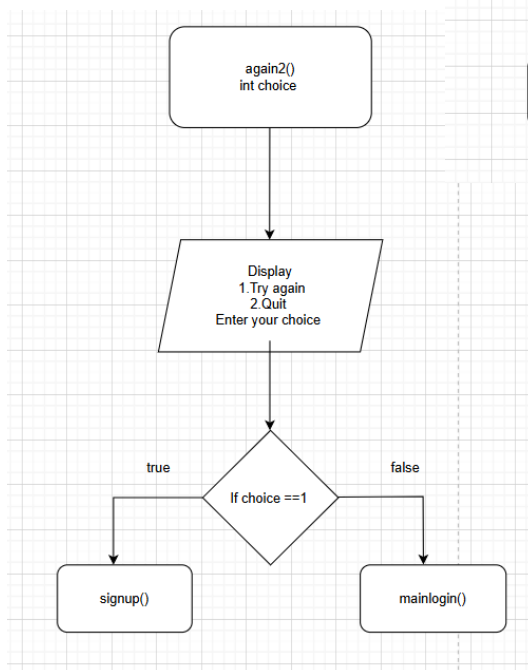
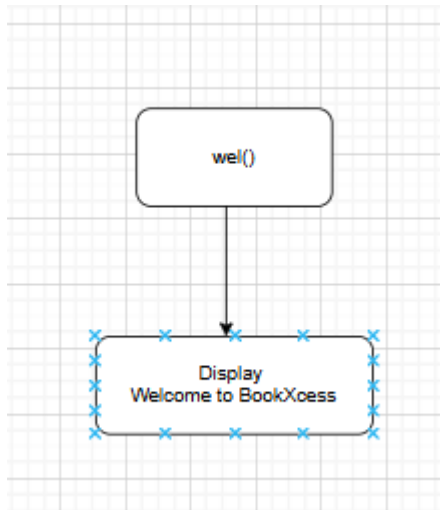
cin>>book.price ;
cout<<"\t+-----+<<endl;

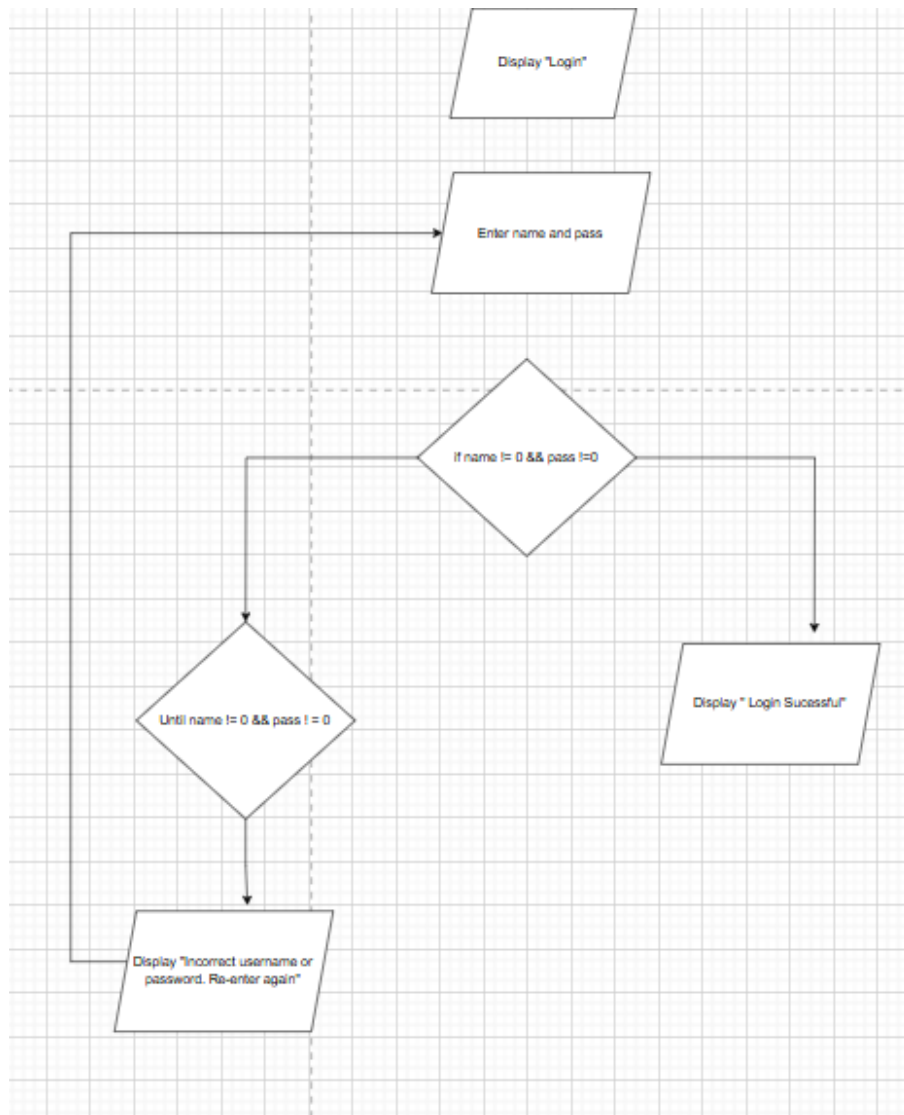
romantic<<book.title<<"\n";
romantic<<book.author<<"\n";
romantic<<book.price<<"\n";
romantic.close();
break;

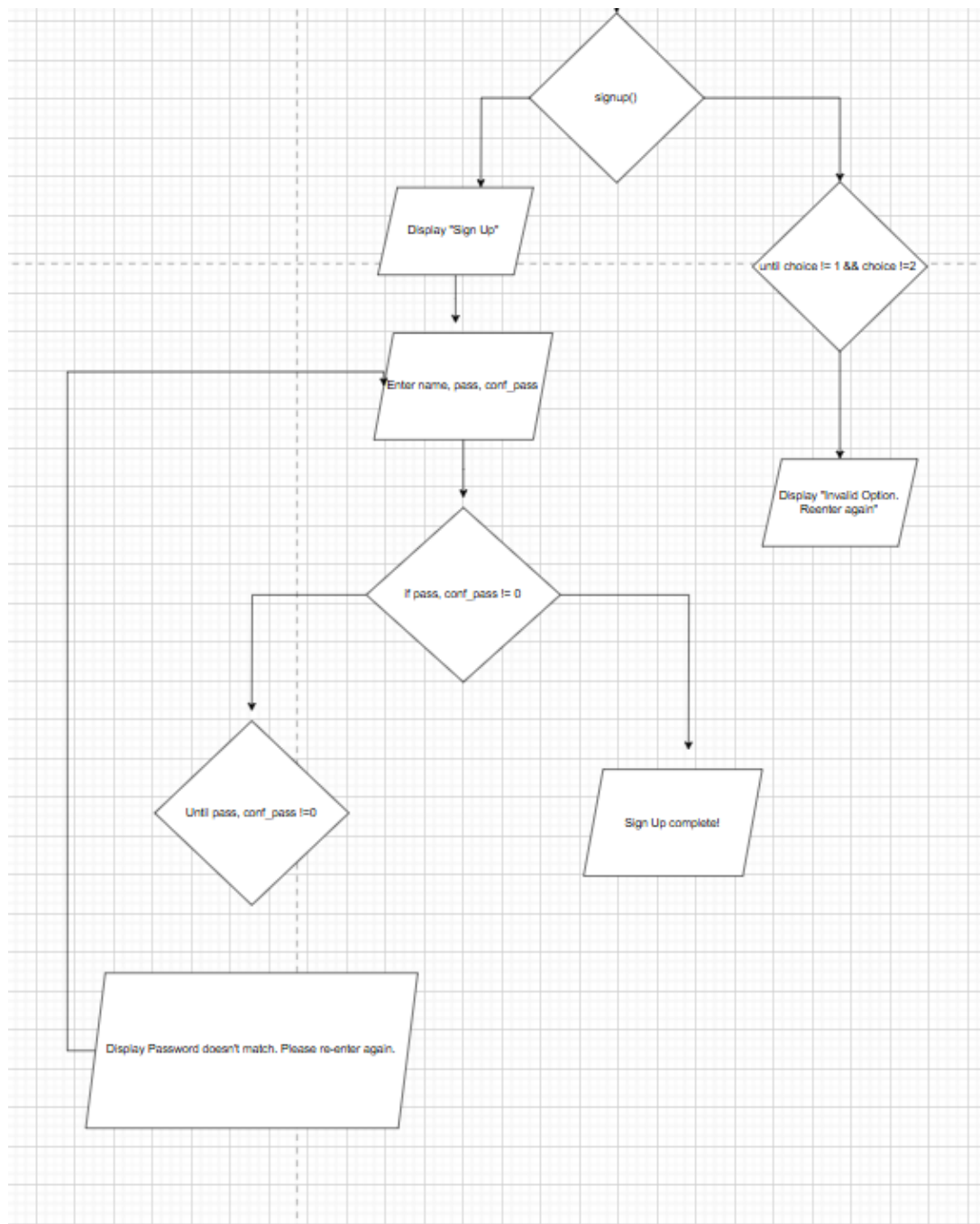
    }
    again3();
}
else
{
    again();
    addbook();
}
}

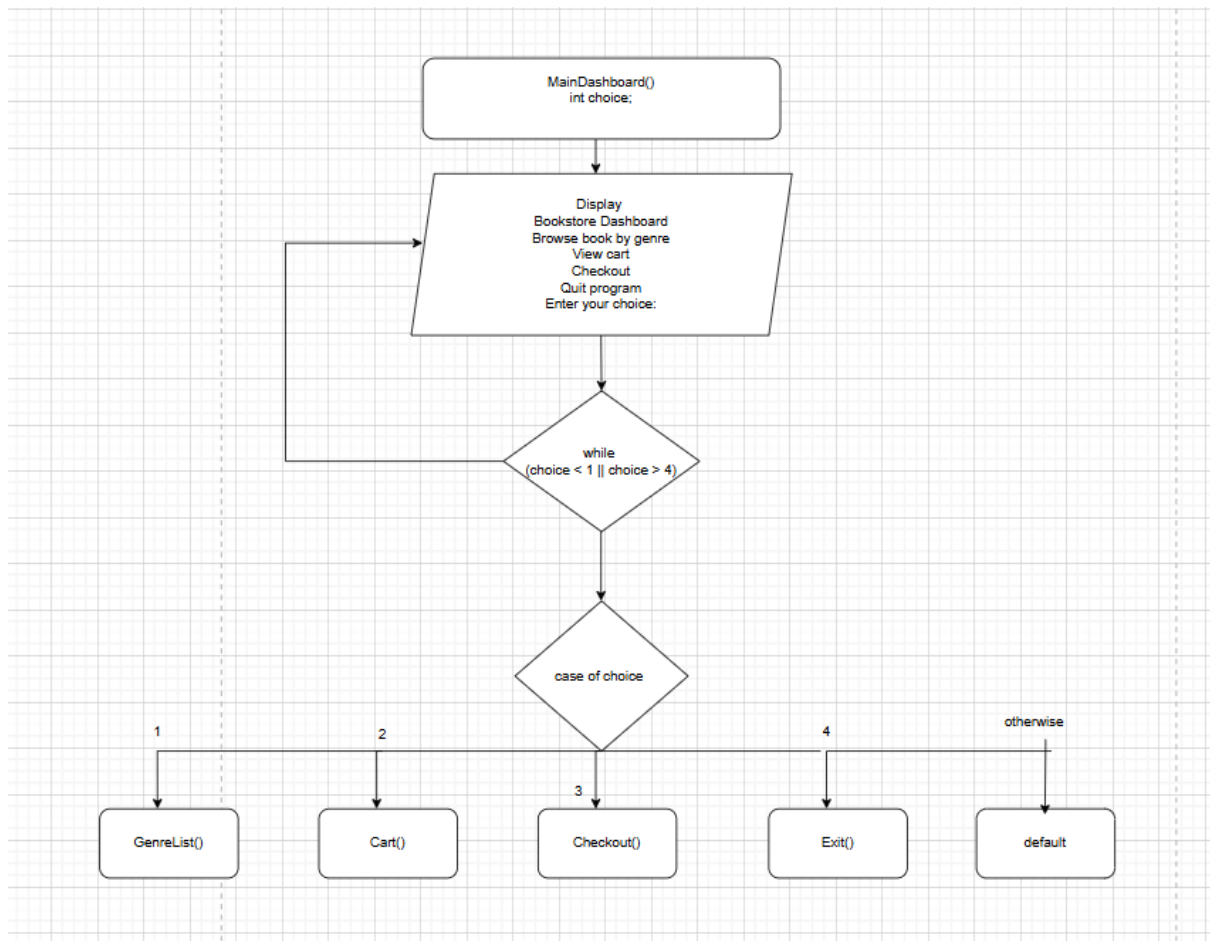
```

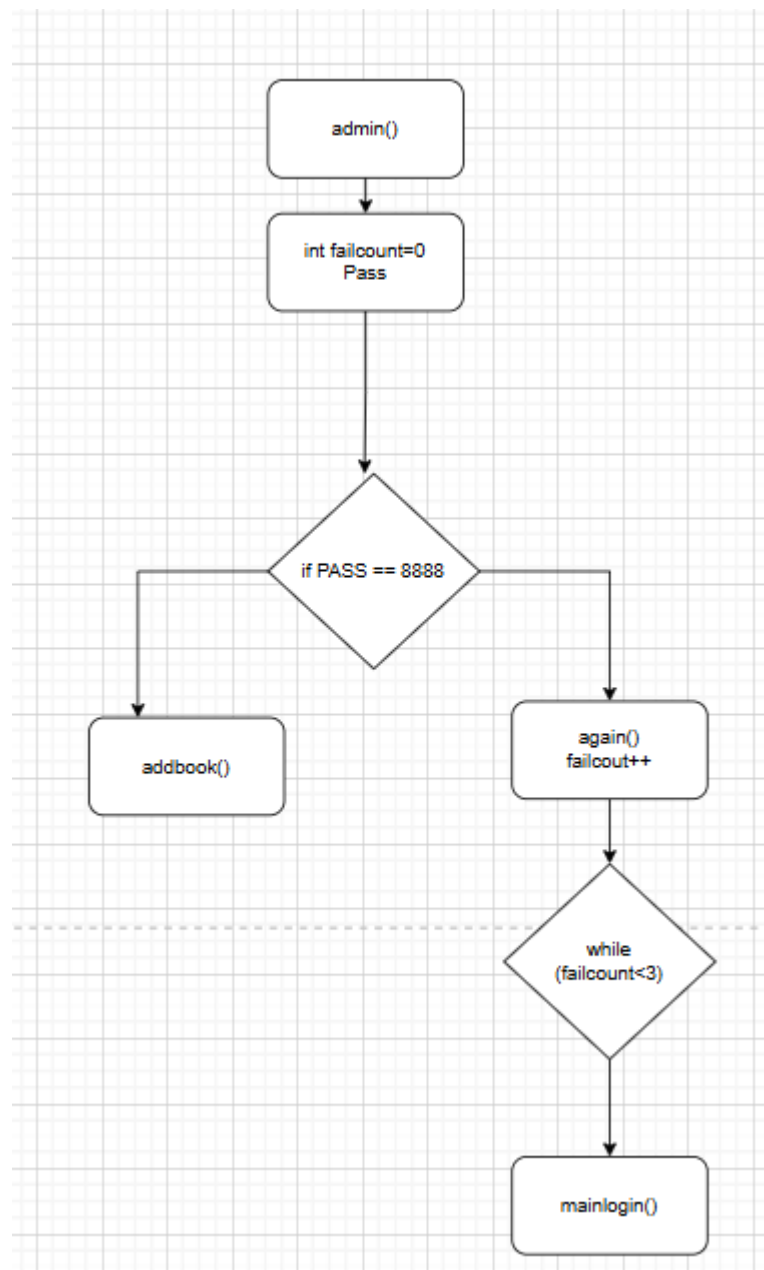


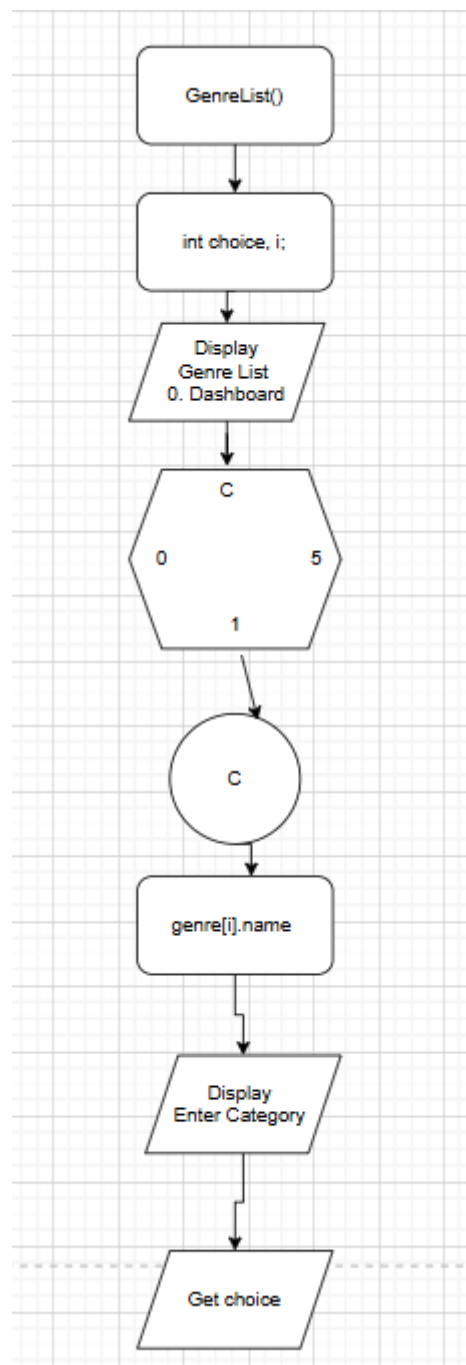


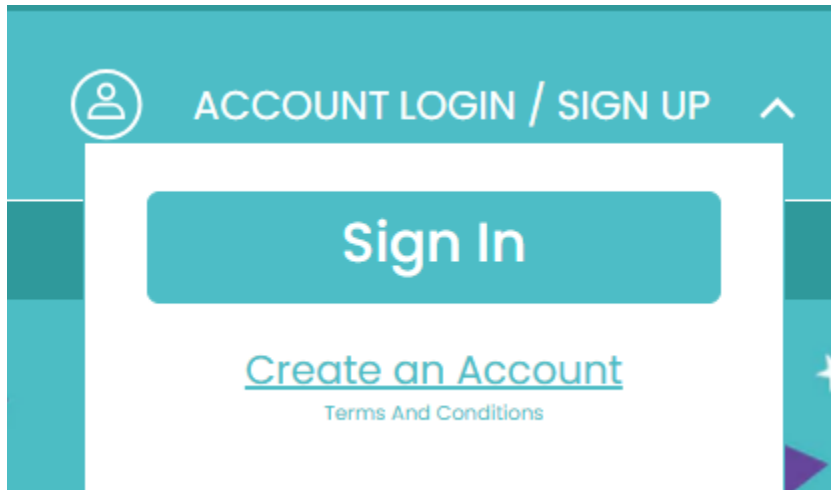
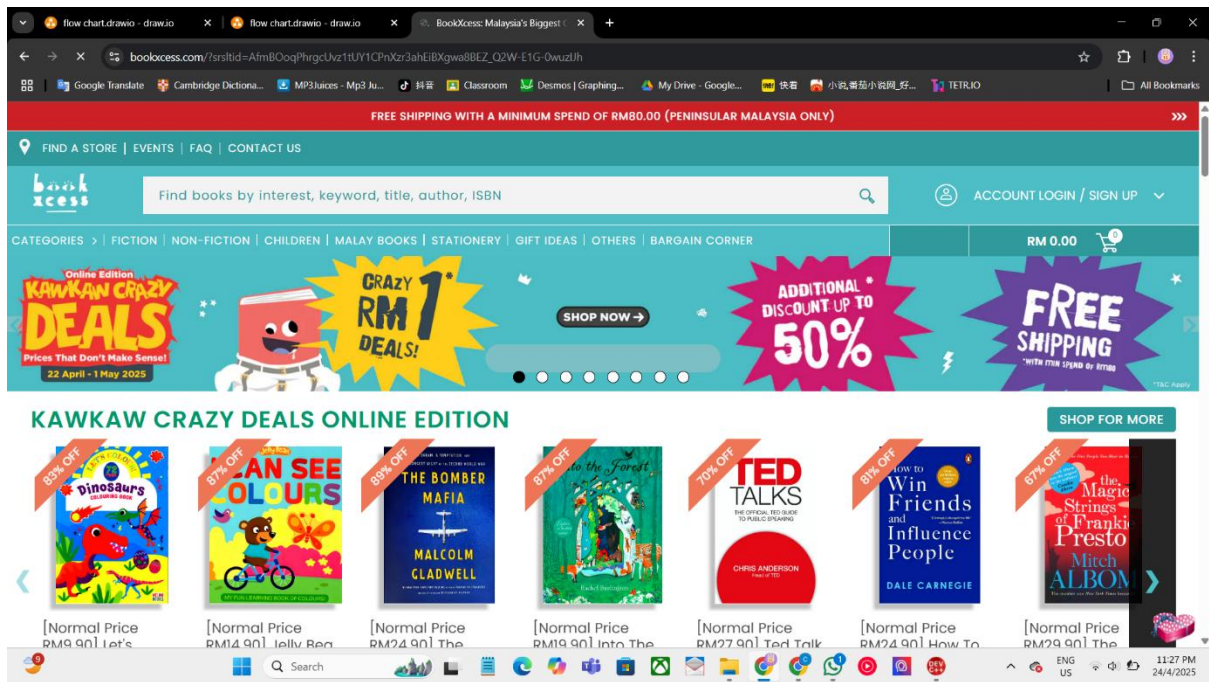












Create account

Already have an account? [Login here](#)

Salutation*

Miss

First name*

qwert

Last name*

qwer

Nationality*

Malaysian

Shipping Address

Address Line 1*

1234

Address Line 2*

12345

Postcode*

12345

City*

asdf

State*

Kuala Lumpur

Country*

Malaysia

Date of Birth

Date

8

Month

4

Year

2015

Mobile Number*

123456789

Email Address*

123456@gmail.com



I Agree to The **Terms & Conditions**

Sign up to be BookXcess member

Best Sellers

Bestselling Books
Must Read Books
Premium Books

Bookish Treasure

Hidden Gems
Bargain Corner
Boxsets

Gifts & Stationery

Educational Items
Games, Puzzle & Toys
Pop Arts

Merchandise

Children's Books

Activity Books
Board Books
Colouring Books
Comics
Picture Books
Pop Up Books
Reading Books
Religious Books
Reference Books
Sound Books
Sticker Books
Workbooks
Frozen

View All

Non-Fiction

Architecture & Design
Arts & Photography
Baking
Beauty & Fashion
Biographies & Memoirs
Business & Economics
Crafts & Hobbies
Cookbooks, Food & Drinks
Entertainment
Graphics & Designs
Health & Wellness
Healthy Cooking
History & Politics

Personal Development

Pets & Nature
Puzzles & Games
References
Relationship & Family
Sports
Transportation
Travel
Spirituality & Religion
View All
Science & Technology
Colouring Books

Fiction

Asian Literature
Contemporary Fiction

Graphic Novels And Comics

Historical Fiction
Romance
Young Adults
View All

Languages

Malay

Buku Kanak -Kanak
Buku Latihan & Rujukan
Novel / Fiksyen
Bacaan Umum / Bukan
Fiksyen

Buku Agama

Chinese

儿童图书 (Children Books)

IN YOUR CART >



Sourdough

RM 119.80

- 2 +



ADD TO WISHLIST



REMOVE ITEM

TOTAL CART VALUE

RM 119.80

CHECK OUT

CONTINUE SHOPPING

CART SUMMARY

Subtotal (1 item) RM 174.75

Savings (-RM 114.85)

Total Cart Value **RM 59.90**

Welcome to BookXcess!

1. Login

2. Sign Up

3. Admin

Enter 1 to Login. Enter 2 to Sign Up. Enter 3 to Admin.
Enter your choice : 1

Login

Username : qwrt
Password : abc1234
Login Failed !
Incorrect username or password.Re-enter again.
Username : qwert
Password : abc1234

Login successful!

Bookstore Dashboard

1. Browse book by genre
2. View cart
3. Checkout
4. Quit program

Enter your choice: 4

Exiting program...
Thank you for visiting! Goodbye.

Welcome to BookXcess!		
1. Login	2. Sign Up	3.Admin
Enter 1 to Login. Enter 2 to Sign Up. Enter 3 to Admin. Enter your choice : 2		
Sign Up		
Username:qwert Password:asdfg Confirm Password:zxcvb Sign Up Failed ! Password doesn't match. Please re-enter again. Password:1234 Confirm Password:1234		
Sign up complete!		
Bookstore Dashboard		
1. Browse book by genre 2. View cart 3. Checkout 4. Quit program		
Enter your choice: 4		
Exiting program... Thank you for visiting! Goodbye.		

```

+-----+
|                                     |
|                               Welcome to BookXcess!                        |
|                                     |
+-----+-----+-----+
|                                     |
|      1. Login          |      2. Sign Up          |      3.Admin          |
|                                     |
+-----+-----+-----+
|                                     |
|           Enter 1 to Login. Enter 2 to Sign Up. Enter 3 to Admin.        |
|       Enter your choice : 3                                             |
|                                     |
+-----+-----+-----+
|                                     |
|                                Admin                                       |
|                                     |
+-----+-----+-----+
|Admin Code:1234|
|                                     |
+-----+-----+-----+
|                                     |
|                   Invalid Option. Reenter again                         |
|                                     |
+-----+-----+-----+
|Admin Code:8888|
|                                     |
+-----+-----+-----+
|                                     |
|                           CHOOSE CATEGORY                              |
|                                     |
+-----+-----+-----+
|                                     |
|      1. Baking              |      2. Business & Economics            |
|      3. Crime, Thriller & Mystery                    |
|      4. Fantasy & Science Fiction                     |
|      5. Romance             |
|                                     |
+-----+-----+-----+
|Enter your choice: 1|
|                                     |
+-----+-----+-----+
|                                     |
|      Book Title: qwert      |
|      Author: asdfg         |
|      Price: 12345          |
|                                     |
+-----+-----+-----+
|                                     |
|      1. Continue add book   |      2. Quit                       |
|                                     |
+-----+-----+-----+
|                                     |
|           Enter 1 to Continue add book. Enter 2 to Quit.                |
|       Enter your choice : 2                                             |
|                                     |
+-----+-----+-----+
|                                     |
|                               Welcome to BookXcess!                        |
|                                     |
+-----+-----+-----+
|                                     |
|      1. Login          |      2. Sign Up          |      3.Admin          |
|                                     |
+-----+-----+-----+
|                                     |
|           Enter 1 to Login. Enter 2 to Sign Up. Enter 3 to Admin.        |
|       Enter your choice : |

```



```

+-----+
|                                     |
|                                     | Genre List                             |
|                                     |
+-----+
0. Dashboard
1. Romance
2. Science Fiction
3. Thriller
4. Baking
5. Business
+-----+
Enter category: 2
+-----+
+-----+ Science FictionBooks +-----+
+-----+
1.      Dune
  Author : Frank Herbert
  Price  : RM  50.5
+-----+
2.      Red Rising
  Author : Pierce Brown
  Price  : RM  69.9
+-----+
3.      Star Maker
  Author : Olaf Stapledon
  Price  : RM  70.5
+-----+
4.      The Calculating Stars
  Author : Mary Robinette Kowal
  Price  : RM  67.8
+-----+
5.      Ready Player One
  Author : Ernest Cline
  Price  : RM  87.7
+-----+
6. Go back to genre list
+-----+

Enter the book number to add to cart: 1
+-----+

'Dune' added to cart!
+-----+

Add another book from the same genre? (Y/N): y
+-----+
+-----+ Science FictionBooks +-----+
+-----+
1.      Dune
  Author : Frank Herbert
  Price  : RM  50.5
+-----+
2.      Red Rising
  Author : Pierce Brown
  Price  : RM  69.9
+-----+
3.      Star Maker
  Author : Olaf Stapledon
  Price  : RM  70.5
+-----+
4.      The Calculating Stars
  Author : Mary Robinette Kowal
  Price  : RM  67.8
+-----+
5.      Ready Player One
  Author : Ernest Cline
  Price  : RM  87.7
+-----+
6. Go back to genre list
+-----+

Enter the book number to add to cart: |

```

+-----+
| Your Cart |
+-----+

- 1. Dune - RM50.5
- 2. Star Maker - RM70.5

Total: RM121

- +-----+
- | |
|-----------------------|
| 1.Proceed to checkout |
| 2.Remove item |
| 3.Return to dashboard |
- +-----+

Enter your choice: 2
Enter the number of the item to remove: 1

+-----+
| Item removed successfully. |
+-----+

+-----+
| Your Cart |
+-----+

- 1. Star Maker - RM70.5

Total: RM70.5

- +-----+
- | |
|-----------------------|
| 1.Proceed to checkout |
| 2.Remove item |
| 3.Return to dashboard |
- +-----+

Enter your choice: 1

Items in your cart:
1. Star Maker - RM 70.5
Shipping cost: RM5
Total (incl. shipping): RM75.5

- +-----+
- | | |
|--------------------|-------------------|
| 1. Proceed Payment | 2. Return to cart |
|--------------------|-------------------|
- +-----+

Enter your choice : |

1. Proceed Payment	2. Return to cart
--------------------	-------------------

Enter your choice : 1

Checkout Please enter your delivery details below.

Full Name : qwe
Address Line : qwe
City : qwe
Postcode : 12
Phone number(without special character): 12

Confirm Checkout?	
1. Confirm and Pay	2. Cancel

Enter your choice: 1

Payment successful!

Do you want back to dashboard? (Press 'y' to Yes / Press others to No): |

baking.txt

Flour Water Salt Yeast

Ken Forkish

76.30

Dessert Person

Claire Saffitz

56.90

Baking Across America

Join B. Dylan Hollis

83.20

Marry Berry's Baking Bible

Marry Berry

63.80

The Elements of Dessert

Francisco Migoya

92.50

Business.txt

The Lean Startup

Eric Ries

83.40

The 7 Habits of Highly Effective People

Stephen R. Covey

77.90

Rich Dad Poor Dad

T. Kiyosaki

64.70

Deep Work

Dale Carnegie

81.00

Never Split the Difference

Chriss Voss

56.90

romance.txt

The Love Hypothesis

Ali Hazelwood

72.80

Book Lovers

Emily Henry

57.90

It Ends with Us

Colleen Hoover

85.90

The Hating Game

Lucy Hale

52.90

The Notebook

Nicholas Sparks

42.80

scifi.txt

Dune

Frank Herbert

50.50

Red Rising

Pierce Brown

69.90

Star Maker

Olaf Stapledon

70.50

The Calculating Stars

Mary Robinette Kowal

67.80

Ready Player One

Ernest Cline

87.70

thriller.txt

Verity

Colleen Hoover

57.80

The Housemaid's Secret

Freida McFadden

69.40

The Quiet Girl

S.F. Kosa

78.50

The Perfect Nanny

Leila Slimani

68.60.

My Sister, the Serial Killer

Oyinkan Braithwaite

87.30